

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

from sklearn.datasets import fetch_california_housing

%matplotlib inline

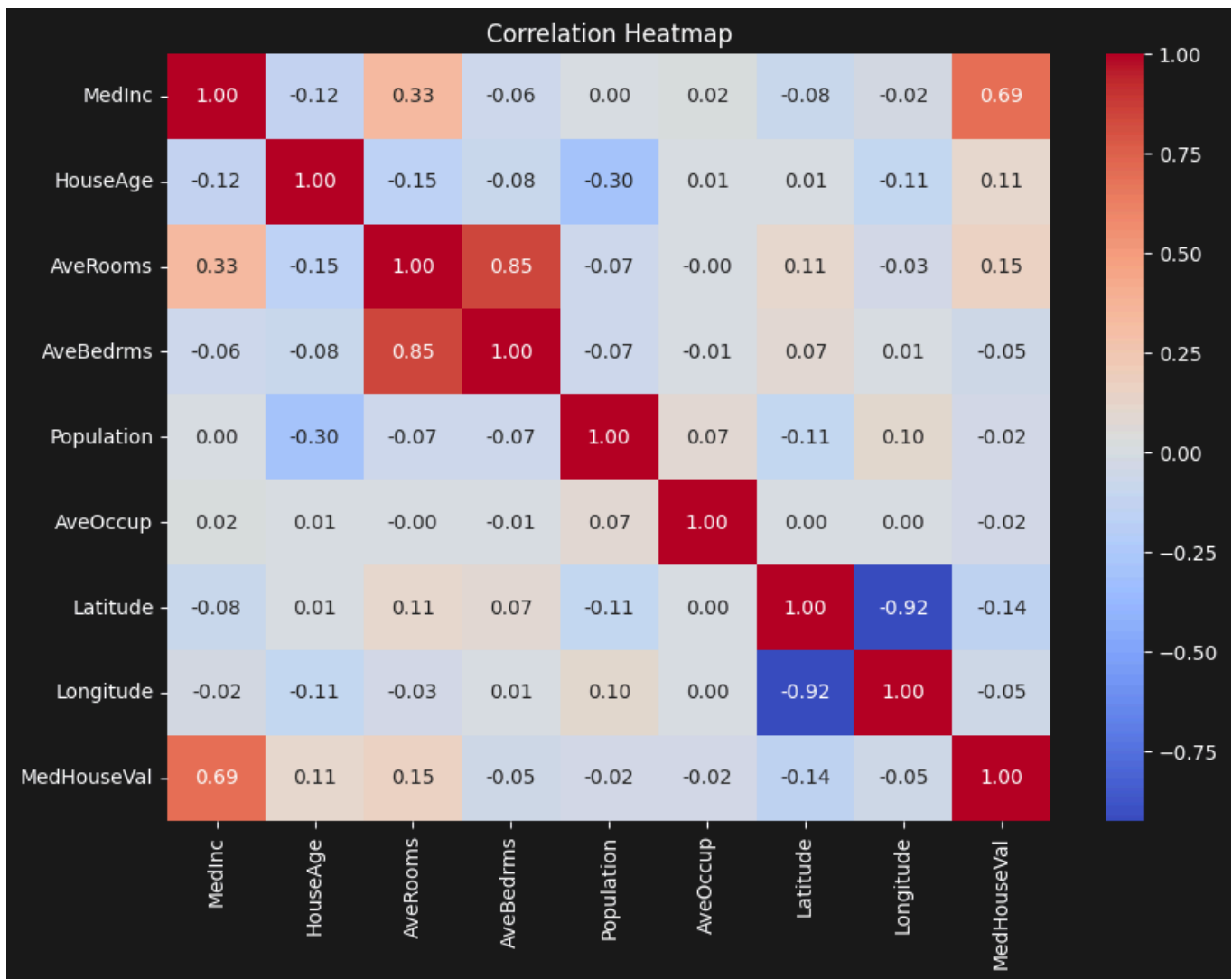
housing = fetch_california_housing(as_frame=True)

df = housing.frame
df.head()
```

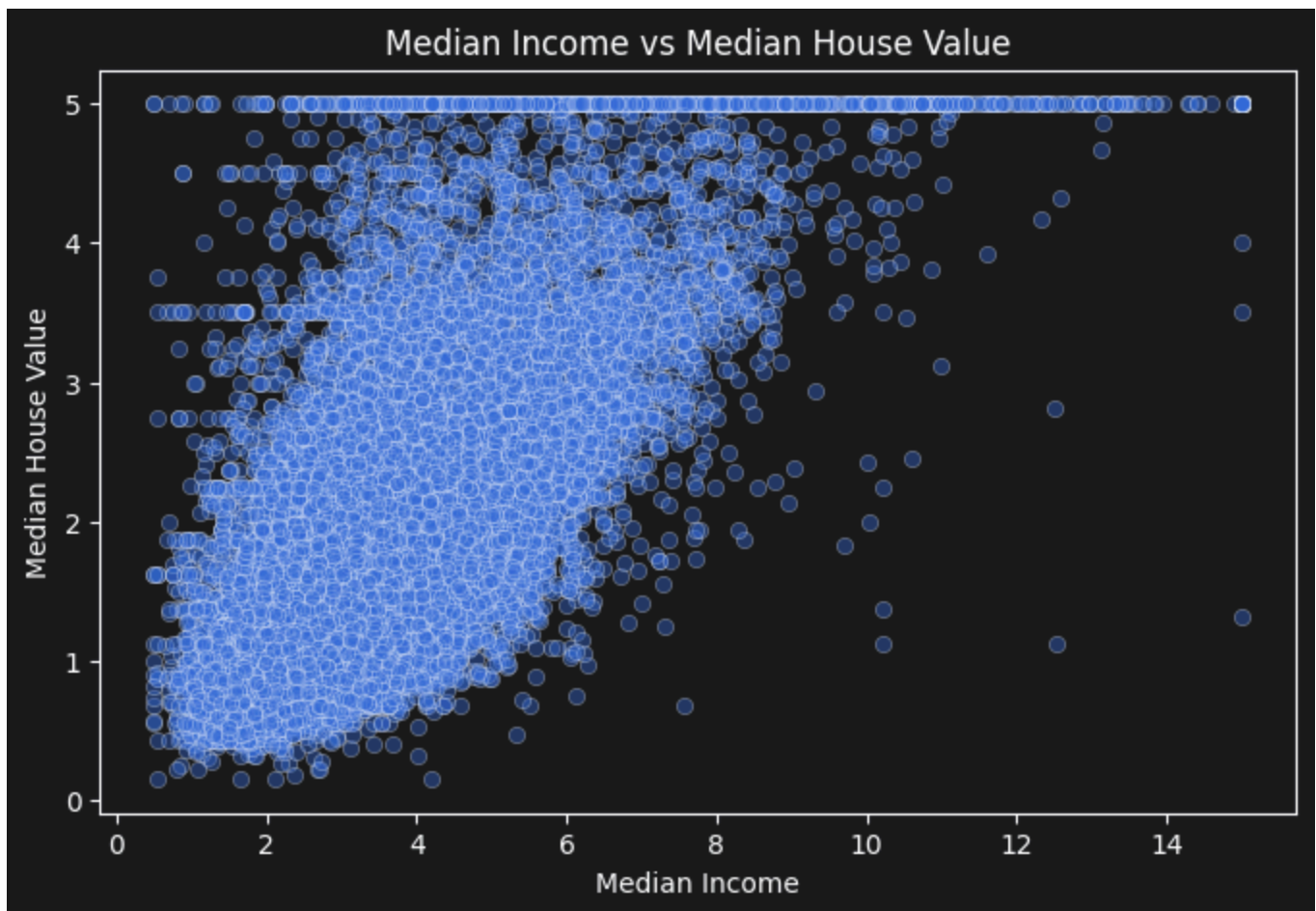
```
X = housing.data
y = housing.target
```

```
df.columns
```

```
plt.figure(figsize=(10, 7))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Correlation Heatmap")
plt.show()
```



```
plt.figure(figsize=(8, 5))
sns.scatterplot(data=df, x="MedInc", y="MedHouseVal", alpha=0.4)
plt.title("Median Income vs Median House Value")
plt.xlabel("Median Income")
plt.ylabel("Median House Value")
plt.show()
```



```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score

%matplotlib inline
X = df[["MedInc"]]
y = df["MedHouseVal"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

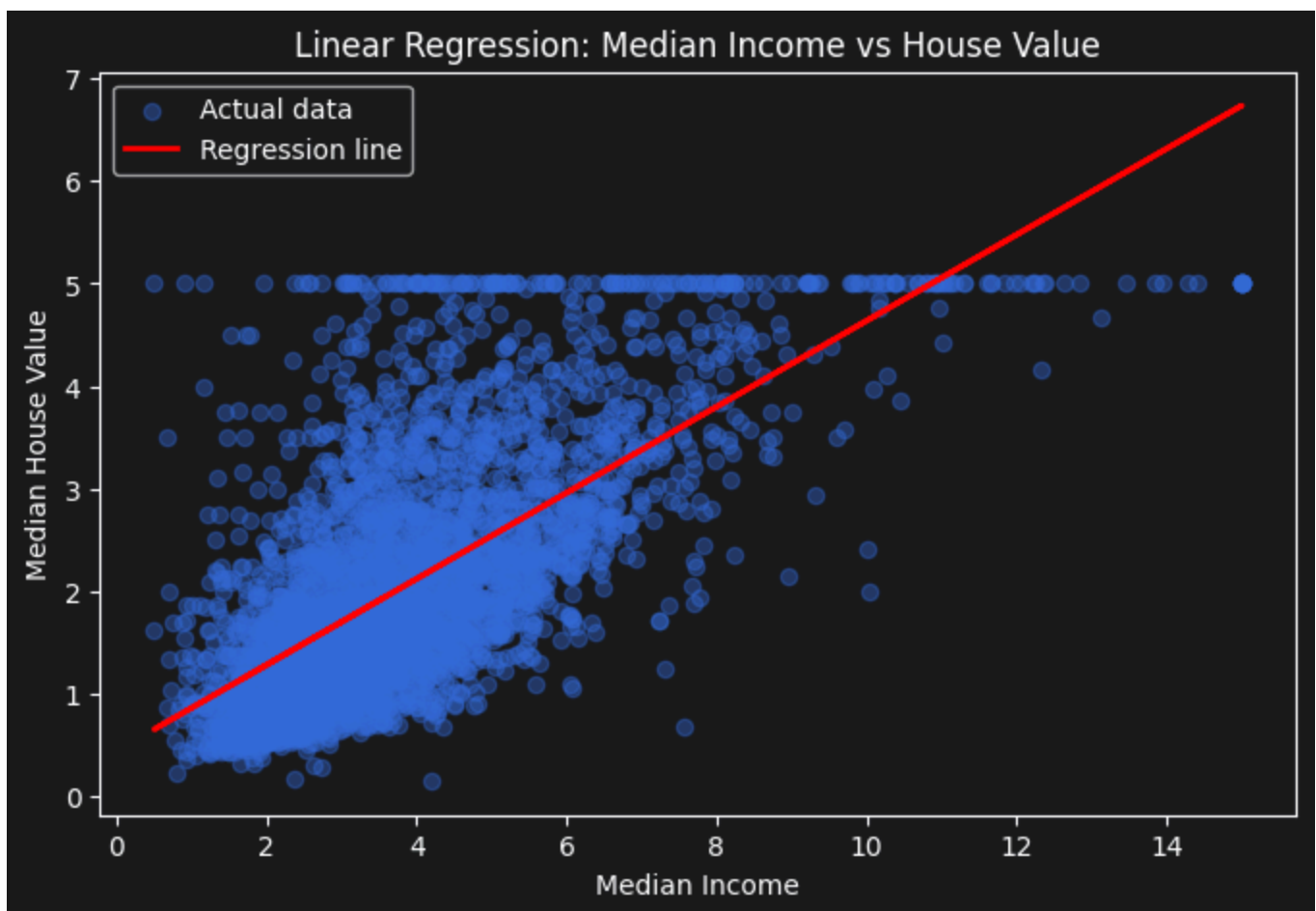
model = LinearRegression()
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("Intercept:", model.intercept_)  
print("Coefficient:", model.coef_[0])
```

```
print(f"MedHouseVal = {model.intercept_:.2f} + {model.coef_[0]:.2f} *  
MedInc")
```

```
plt.figure(figsize=(8, 5))  
  
plt.scatter(X_test, y_test, alpha=0.4, label="Actual data")  
plt.plot(X_test, y_pred, color="red", linewidth=2, label="Regression line")  
  
plt.title("Linear Regression: Median Income vs House Value")  
plt.xlabel("Median Income")  
plt.ylabel("Median House Value")  
plt.legend()  
plt.show()
```



```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

```
print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
X = df.drop("MedHouseVal", axis=1)
y = df["MedHouseVal"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
multi_model = LinearRegression()
multi_model.fit(X_train, y_train)
```

```
y_pred_multi = multi_model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred_multi)
mse = mean_squared_error(y_test, y_pred_multi)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred_multi)
```

```
print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```